- It contains information required for the formation of proteins.
- It controls the developmental process and life activities of an organism.

18.7 Sex Determination

The formation of zygote into male or female sex during development is called sex determination. Sex is determined by the chromosomes of an individual.

18.7.1 Sex Determination in Human

Recall that human beings have 23 pairs of chromosomes out of which 22 pairs are autosomes and one pair (23rd pair) is the sex chromosome. The female gametes or the eggs formed are similar in their chromosome type (22+XX). Therefore, human females are **homogametic**.

The male gametes or sperms produced are of two types. They are produced in equal proportions. The sperm bearing **(22+X)** chromosomes and the sperm bearing **(22+Y)** chromosomes. The human males are called **heterogametic.**

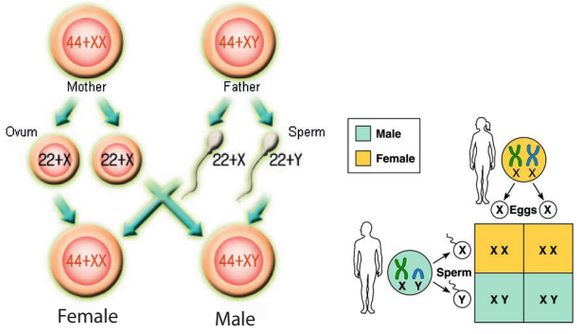


Figure 18.9 Sex determination in human

It is a chance of probability as to which category of sperm fuses with the egg. If the egg (X) is fused by the X-bearing sperm an **XX individual (female)** is produced. If the egg (X) is fused by the Y-bearing sperm an **XY individual (male)** is produced. The sperm, produced by the father, determines the sex of the child. The mother is not responsible in determining the sex of the child.

Now let's see how the chromosomes take part in this formation. Fertilization of the egg (22+X) with a sperm (22+X) will produce a female child (44+XX). while fertilization of the egg (22+X) with a sperm (22+Y) will give rise to a male child (44+XY).

18.7.2 Mutation

The term mutation was introduced by **Hugo De Vries** in 1901 when he observed phenotypic changes in the evening primrose plant, *Oenothera lamarckiana*. Mutation is an inheritable sudden change in the genetic material (DNA) of an organism. Mutations are classified into two main types, namely chromosomal mutation and gene mutation.

1. **Chromosomal mutation**

The **sudden change** in the **structure** or **number of chromosomes** is called chromosomal mutation. This may result in

(i) Changes in the structure of chromosomes: Structural changes in the chromosomes usually occurs due to errors in cell division. Changes in the number and arrangement of genes takes place as a result of deletion, duplication, inversion and translocation in chromosomes.

(ii) Changes in the number of chromosomes: They involve addition or deletion in the number of chromosomes present in a cell. This is called **ploidy**. There are two types of ploidy

(a) Euploidy (b) Aneuploidy.

Euploidy: It is the condition in which the individual bears **more than the usual number** of diploid (2n) chromosomes. If an individual has three haploid sets of chromosomes, the condition is called **triploidy** (3n). Triploid plants and animals are typically sterile.If it has four haploid sets of chromosomes, the condition is called **tetraploidy** (4n). Tetraploid plants are advantageous as they often result in increased fruit and flower size.

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